INCH-POUND

MIL-P-44151A
30 September 1992
SUPERSEDING
MIL-P-44151(GL)
30 June 1985

MILITARY SPECIFICATION

PACKAGING INTO POUCHES AND MEAL ASSEMBLY OF ORAL LIQUID FEEDING POWDERS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers the packaging and meal assembly of food product powder in pouches for use in oral liquid feeding. The products are unit packed as meals that provide a nutritionally adequate diet, when consumed as in the suggested meals.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development and Engineering Center, Natick, MA 01760-5018 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8970

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SPECIFICATIONS

FEDERAL

L-P-378	-	Plastic Sheet and Strip, Thin Gauge, Polyolefin
QQ-A-1876	-	Aluminum Foil
PPP-B-26	-	Bags, Plastic, (General Purpose)
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-L-35078	-	Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For
MIL-S-44170	-	Spaghetti and Meat with Sauce, Powder (For Oral Liquid Feeding)
MIL-C-44174	-	Chili Con Carne, Powder (For Oral Liquid Feeding)
MIL-C-44175	-	Corn, Buttered, Powder (For Oral Liquid Feeding)
MIL-H-44176	-	Hominy Grits with Cheese, Powder (For Oral Liquid Feeding)
MIL-P-44182	-	Potatoes, Mashed, Powder (For Oral Liquid Feeding)
MIL-D-44414	-	Dairyshake Powder
MIL-C-44437	-	Cheesecake, Powder (For Oral Liquid Feeding)
MIL-C-44438	-	Corn and Potatoes, Creamed, Powder (For Oral Liquid Feeding)
MIL-G-44439	-	Gingerbread, Powder (For Oral Liquid Feeding)
MIL-P-44440	-	Potatoes, Delmonico, Powder (For Oral Liquid Feeding)
MIL-S-44441	-	Squash, Buttered, Powder (For Oral Liquid Feeding)
MIL-C-44442	-	Cake, Blueberry Upside-Down, Powder (For Oral Liquid Feeding)
MIL-B-44443	-	Beans, Baked, Powder (For Oral Liquid Feeding)
MIL-S-44444	-	Strawberry Shortcake, Powder (For Oral Liquid Feeding)
MIL-A-44445	-	Apple Brown Betty, Powder (For Oral Liquid Feeding)
MIL-B-44446	-	Banana-Strawberry Breakfast Drink, Powder (For Oral Liquid Feeding)
MIL-C-44447	-	Cauliflower and Cheese, Powder (For Oral Liquid Feeding)
MIL-E-44448	-	Escalloped Potatoes and Ham, Powder (For Oral Liquid Feeding)
MIL-H-44449	-	Ham and Cheese Souffle, Powder (For Oral Liquid Feeding)
MIL-O-44450	-	Oatmeal with Maple and Brown Sugar, Powder (For Oral Liquid Feeding)
MIL-P-44451	-	Pea Casserole, Powder (For Oral Liquid Feeding)
MIL-T-44452	-	Taco Grande, Powder (For Oral Liquid Feeding)

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MIL-O-44453 - Omelet, Green Pepper and Ham, Powder (For Oral Liquid Feeding)
MIL-C-44454 - Chicken Barbecue, Powder (For Oral Liquid Feeding)
MIL-H-44455 - Ham and Cheese Fondue, Powder (For Oral Liquid Feeding)
MIL-P-44456 - Pork Oriental and Vegetables with Noodles, Powder (For Oral Liquid Feeding)
MIL-R-44457 - Rice, Fried, Powder (For Oral Liquid Feeding)
MIL-F-44459 - Farina with Peaches and Cream, Powder (For Oral Liquid Feeding)
MIL-G-44460 - Green Bean Casserole, Powder (For Oral Liquid Feeding)
MIL-P-44461 - Potatoes, Lyonnaise, Powder (For Oral Liquid Feeding)
MIL-C-44462 - Cake, Chocolate Mocha, Powder (For Oral Liquid Feeding)

STANDARDS

FEDERAL

FED-STD-595 - Colors Used in Government Procurement

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder (21 CFR Parts 1-199)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

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2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

F 88 - Seal Strength of Flexible Barrier Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., Traffic Department, 1616 P Street NW, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Suite 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Components and materials.

3.1.1 Food product. The food product to be unit packed and assembled shall be as specified in the applicable document (see 3.4).

3.1.2 Pouch.

3.1.2.1 Pouch material. The pouch material shall be a 3-ply laminate consisting of 0.0020-inch thick ionomer or polyethylene film laminated or extrusion-coated to 0.00035-inch thick aluminum foil, which is then laminated

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to 0.00050-inch thick polyester. The three plies shall be laminated so that the aluminum foil is between the other two layers. The pouch shall be formed with the polyester on the exterior. The pouch material shall be approved for food use and shall show no evidence of delamination or degradation when heat sealed or fabricated into pouches. The complete exterior surface of the pouch shall be uniformly colored in the range of green 34079 through 34087 or 24052 through 24087, or brown 30045 through 30118 (excluding 30109) or 10045 of FED-STD-595.

3.1.2.2 Pouch requirements. The size and design of the pouch shall be as specified in Figure 1. Pouch side and bottom seams shall be heat sealed in the locations shown in Figure 1 and shall be such that the heat seal shall conform to seal strength and vacuum retention requirements in 3.2.

3.2 Pouch filling and sealing. Pouches shall be filled in accordance with the net weight requirements of the applicable product document. The filling operation shall be carefully executed to prevent contamination of the closure seal interface surfaces by contact with the product, splashing, or transfer from the filler nozzle. Powdered product must be distributed uniformly throughout pouch so as not to exceed 1/2 inch in thickness. The pouch shall be formed by heat sealing and provided with tear notches as shown in Figure 1. One side of the open end of the pouch may be provided with an extended or fold-over lip (see Figure 1). The filled pouch shall be closed by evacuating the air within a sealing chamber under a vacuum of not less than 28 inches of mercury. The pouch shall be closed at this vacuum level with a continuous heat seal not less than 1/4 inch wide extending across the complete width of the pouch. If thermal impulse or combination (heated curved bar with thermal impulse) sealing is used, any seal width from 1/8 to 7/16 inch will be acceptable. The closure seal location shall be as shown in Figure 1. Both the pouch manufacturer's heat seals and the closure heat seal shall have an average seal strength of 6 pounds per inch of width and each individual test specimen shall have a seal strength of not less than 5 pounds per inch width when tested in accordance with 4.2.1.2.1 and 4.2.5.

3.3 Pouch labeling and marking. Pouches shall be labeled in a permanent, contrasting color directly on the packaging material in a manner that does not damage the pouches with a food compatible ink which is free of carcinogenic elements or ingredients. No marking shall be closer than 1/16 inch to any seal. Pressure-sensitive labels may be used instead of marking directly on the packaging material. Labels shall be of a water-resistant grade of paper or film, flat-back, coated on one side with pressure-sensitive adhesive. The texture of the material shall permit flexibility. Labels shall be suitable for printing or writing on with ink without feathering or spreading. The adhesive shall be of a pressure-sensitive permanent type. The adhesive shall be water-insoluble, homogeneous, and shall be coated in a smooth layer on one side of the label. The adhesive shall require no solvent, heat, or other preparation prior to application. Adhesive shall be of a type that will adhere to the pouch surface under high or low

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temperatures. Dimensions of the label shall be not greater than the dimensions of the surface of the product where the label is applied on the pouch. Ink and adhesive used must be compatible for use with food products and shall be free of carcinogenic elements or ingredients. Letter size shall be not less than 5/32 inch high. The pouch label shall contain the following information:

PRODUCT NAME

INGREDIENTS (to be listed in accordance with the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder)

DIRECTIONS FOR USE (as specified below)

NET WEIGHT

Contractor's name and address

Date of pack 1/

1/ Each pouch shall have the date of pack noted by using a four digit code beginning with the final digit of the current year followed by the three digit Julian day code. (For example, March 19, 1991 would read 1079.) The Julian day code shall represent the date the product is unit packed into the pouch.

a. For Apple Brown Betty; Banana-Strawberry Breakfast Drink; Cake, Blueberry Upside Down; Cake, Chocolate Mocha; Cheesecake; Gingerbread; and Strawberry Shortcake; the following directions for use shall be specified:

DIRECTIONS FOR USE:

1. Empty contents of pouch into a food blender.
2. Add 6 ounces cold water or milk.
   (For a pureed diet, use 3 ounces of liquid.)
3. Blend for 30 seconds on high speed.
4. Consume products within 2 hours of preparation or refrigerate.

b. For Chicken Barbecue; Chili Con Carne; Omelet, Green Pepper and Ham; Ham and Cheese Fondue; Ham and Cheese Souffle; Hominy Grits with Cheese; Pork Oriental and Vegetables with Noodles; Spaghetti and Meat with Sauce; and Taco Grande; the following directions for use shall be specified:

DIRECTIONS FOR USE:

1. Empty contents of pouch into a food blender.
2. Add 6 ounces hot water or milk (not less than 140°F).
   (For a pureed diet, use 3 ounces of liquid.)
3. Blend for 30 seconds on high speed.
4. Consume products within 2 hours of preparation or refrigerate.

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c. For Beans, Baked; Cauliflower and Cheese; Corn, Buttered; Corn and Potatoes, Creamed; Escalloped Potatoes and Ham; Farina with Peaches and Cream; Green Bean Casserole; Oatmeal with Maple and Brown Sugar; Pea Casserole; Potatoes, Delmonico; Potatoes, Lyonnaise; Potatoes, Mashed; Rice, Fried; and Squash, Buttered; the following directions for use shall be specified:

DIRECTIONS FOR USE:

1. Empty contents of pouch into a food blender.
2. Add 8 ounces hot water or milk (not less tha 140°F). (For a pureed diet, use 4 ounces of liquid.)
3. Blend for 30 seconds on high speed.
4. Consume products within 2 hours of preparation or refrigerate.

3.4 Meal components. The components required for meal assembly shall be as specified in table I.

TABLE I. Meal components

Components	Document reference	Used in meal
Apple Brown Betty	MIL-A-44445	Day 1, Lunch
Beans, Baked	MIL-B-44443	Day 2, Dinner
Banana-Strawberry Breakfast Drink	MIL-B-44446	Day 1, Breakfast
Cake, Blueberry Upside-Down	MIL-C-44442	Day 1, Dinner
Cake, Chocolate Mocha	MIL-C-44462	Day 2, Dinner
Cauliflower and Cheese	MIL-C-44447	Day 1, Dinner
Cheesecake	MIL-C-44437	Day 3, Dinner
Chicken Barbecue	MIL-C-44454	Day 2, Dinner
Chili Con Carne	MIL-C-44174	Day 1, Dinner
Corn, Buttered	MIL-C-44175	Day 2, Dinner
Corn and Potatoes, Creamed	MIL-C-44438	Day 3, Dinner
Dairyshake, Banana	MIL-D-44414	Day 2, Dinner
Dairyshake, Chocolate	MIL-D-44414	Day 1, Dinner and Day 3, Dinner
Dairyshake, Eggnog	MIL-D-44414	Day 2, Lunch
Dairyshake, Orange	MIL-D-44414	Day 2, Breakfast
Dairyshake, Strawberry	MIL-D-44414	Day 1, Lunch and Day 3, Lunch
Dairyshake, Vanilla	MIL-D-44414	Day 1, Breakfast and Day 3, Breakfast
Escalloped Potatoes and Ham	MIL-E-44448	Day 1, Dinner
Farina with Peaches and Cream	MIL-F-44459	Day 3, Breakfast
Gingerbread	MIL-G-44439	Day 3, Lunch
Green Bean Casserole	MIL-G-44460	Day 3, Lunch
Ham and Cheese Fondue	MIL-H-44455	Day 3, Breakfast
Ham and Cheese Souffle	MIL-H-44449	Day 1, Lunch
Hominy Grits with Cheese	MIL-H-44176	Day 2, Breakfast

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TABLE I. Meal components (cont'd)

Components	Document reference	Used in meal
Oatmeal with Maple and Brown Sugar	MIL-O-44450	Day 1, Breakfast
Omelet, Green Pepper and Ham	MIL-O-44453	Day 2, Breakfast
Pea Casserole	MIL-P-44451	Day 1, Lunch
Pork Oriental and Vegetables with Noodles	MIL-P-44456	Day 3, Dinner
Potatoes, Delmonico	MIL-P-44440	Day 2, Lunch
Potatoes, Lyonnaise	MIL-P-44461	Day 3, Lunch
Potatoes, Mashed	MIL-P-44182	Day 1, Lunch
Rice, Fried	MIL-R-44457	Day 3, Dinner
Spaghetti and Meat with sauce	MIL-S-44170	Day 3, Lunch
Squash, Buttered	MIL-S-44441	Day 2, Lunch
Strawberry Shortcake	MIL-S-44444	Day 2, Lunch
Taco Grande	MIL-T-44452	Day 2, Lunch

3.5 Meal assembly. One each of the applicable components of a meal indicated in table II shall be unit packed in the polyethylene meal bag as specified in 5.1.1.1. The product shall be distributed within each filled and sealed meal bag to attain a stack height of not more than 1.5 inches for each complete breakfast meal when the pouches are positioned flat and stacked one over the other in vertical alignment. The stack height for each complete lunch or dinner meal shall not exceed 2.5 inches when arranged as specified above.

TABLE II. Meal contents

Meal	Components
	Day 1
Breakfast	Banana-strawberry breakfast drink Oatmeal with maple and brown sugar Dairyshake, vanilla
Lunch	Ham and cheese souffle Pea casserole Potatoes, mashed Apple brown betty Dairyshake, strawberry
Dinner	Chili con carne Cauliflower and cheese Escalloped potatoes with ham Cake, blueberry upside-down Dairyshake, chocolate

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TABLE II. Meal contents (cont'd)

Meal	Components
	Day 2
Breakfast	Hominy grits with cheese Omelet, green pepper and ham Dairyshake, orange
Lunch	Taco grande Squash, buttered Potatoes, delmonico Strawberry shortcake Dairyshake, eggnog
Dinner	Chicken barbecue Corn, buttered Beans, baked Cake, chocolate mocha Dairyshake, banana
	Day 3
Breakfast	Farina with peaches and cream Ham and cheese fondue Dairyshake, vanilla
Lunch	Spaghetti and meat with sauce Green bean casserole Potatoes, lyonnaise Gingerbread Dairyshake, strawberry
Dinner	Pork oriental and vegetables with noodles Corn and potatoes, creamed Rice, fried Cheesecake Dairyshake, chocolate

3.6 Plant qualification. The product shall be packaged and assembled in establishments meeting the requirements of Title 21, Code of Federal Regulations, Part 110, "Current Good Manufacturing Practice in Manufacturing, Packing or Holding Human Food," and the plant sanitation requirements of the appropriate Government inspection agency.

3.7 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA or other Government inspection agency, shall not relieve the contractor of the obligation and responsibility to deliver a product complying with all the requirements of this specification. The contractor shall ensure product compliance prior to submitting the product to the USDA or other Government inspection agency for any inspection.

4.1.1 Inspection and certification. Product acceptability shall be determined by the USDA or other Government inspection agency. The USDA or other Government inspection agency will determine the degree of inspection and supervision necessary to ensure compliance with the requirements of this specification.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.1.1 Pouch material examination. Material listed below shall be accepted on the basis of the manufacturer's certificate of conformance to the indicated requirements.

Material requirement	Requirement paragraph	Test procedure
Ionomer or polyethylene film thickness	3.1.2.1	As specified in L-P-378, except that a machinist's micrometer may be used provided that its graduations and accuracy conform to the requirements of L-P-378.
Polyester film thickness	3.1.2.1	As above.
Aluminum foil thickness	3.1.2.1	As specified in QQ-A-1876.
Laminated material construction	3.1.2.1	Laboratory evaluation.
Color of laminated material	3.1.2.1	Visual evaluation.

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4.2.1.2 Unfilled pouch certification. A certificate of conformance may be accepted as evidence that unfilled pouches conform to the requirements specified in 3.1.2.2. When deemed necessary by the inspection agency, testing of the unfilled pouches for seal strength of manufacturer's seals shall be as specified in 4.2.1.2.1.

4.2.1.2.1 Unfilled pouch seal strength testing. Eight unfilled pouches shall be selected at random from the lot. The unfilled pouches shall be tested for seal strength in accordance with the procedures specified in ASTM F 88 except that testing speed may be 10 or 12 inches per minute. Three adjacent specimens shall be cut from each of the three sealed sides of each pouch in the sample. The results shall be reported to the nearest 0.1 pound. The average seal strength of each seal shall be calculated by averaging the strengths of the three test specimens cut from that seal. Any test specimen or average seal strength failing to meet the requirements of 3.2 shall be classified as a major defect and shall be cause for rejection of the lot.

4.2.2 In-process examination. In-process examination shall be performed to determine conformance to filling, sealing, and packaging requirements. Any nonconformance revealed by actual examination or by review of records or other valid documents shall be cause for rejection of the involved product.

4.2.3 Filled and sealed pouch examination. Filled and sealed pouches shall be examined for defects as listed in table III. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 0.65 for major defects and 2.5 for minor defects.

TABLE III. Filled and sealed pouch defects

Category		Defect
Major	Minor	
101		Not clean 1/
102		Presence of delamination 2/
103		Closure seal width not as specified
104		Closure seal not located as specified
105		Less than 3/32 inch between inside edge of tear notch and inside edge of seal
106		Labeling missing, incorrect, or illegible
	201	Presence of delamination 2/
	202	Thickness of pouch exceeds 1/2 inch
	203	Tear notches missing or not as specified
	204	Tear notches not located as specified
	205	Depth of tear notches not as specified

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1/ Outer packaging shall be free from foreign matter which is unwholesome, has the potential to cause pouch damage (for example, glass, metal filings, etc.), or generally detracts from the clean appearance of the package. The following examples shall not be scored as defects for unclean:

a. Foreign matter which presents no health hazard or potential pouch damage and which can be readily removed by gently shaking the package or by gently brushing the package with a clean dry cloth.

b. Dried product which effects less than 1/8 of the total surface area of one pouch face (localized and aggregate).

c. Water spots.

d. Very thin film of grease, oil, or product residue which is discernible to touch, but is not readily discernible by visual examination.

2/ Delamination defect classification:

Major - Delamination of the outer ply in the pouch seal area that can be propagated to expose aluminum foil at the food product edge of the pouch after manual flexing of the delaminated area. To flex, the delaminated area shall be held between the thumb and forefinger of each hand with both thumbs and forefingers touching each other. The delaminated area shall be rapidly flexed 10 times by rotating both hands in alternating clockwise-counter-clockwise directions. Care shall be exercised when flexing delaminated areas near the tear notches to avoid tearing the pouch material. After flexing, the separated outer ply shall be grasped between the thumb and forefinger and gently lifted toward the food product edge of the seal or if the separated area is too small to be held between the thumb and forefinger, a number 2 stylus shall be inserted into the delaminated area and a gentle lifting force applied against the outer ply. If separation of the outer ply can be made to extend to the product edge of the seal with no discernible resistance to the gentle lifting, the delamination shall be classified as a major defect. Additionally, spot delamination of the outer ply in the body of the pouch that is able to be propagated beyond its initial borders is also a major defect. To determine if the delaminated area is a defect, use the following procedure: Mark the outside edges of the delaminated area using a bold permanent marking pen. Open the pouch and remove the contents. Cut the pouch transversely not closer than 1/4 inch (plus or minus 1/16 inch) from the delaminated area. The pouch shall be flexed in the area in question using the procedure described above. Any propagation of the delaminated area, as evidenced by the delaminated area exceeding the limits of the outlined borders, shall be classified as a major defect.

Minor - Minor delamination of the outer ply in the pouch seal area is acceptable and shall not be classified as a minor defect unless it extends to within 1/16 inch of the food product edge of the seal. All other minor outer

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ply delamination in the pouch seal area or isolated spots of delamination in the body of the pouch that do not propagate when flexed as described above shall be classified as minor.

4.2.4 Pouch vacuum examination. The filled and sealed pouches shall be visually examined for conformance to the vacuum requirement in 3.2 not less than 96 hours after filling and sealing. The sealed pouch shall continue to exhibit tight adherence to the surface contours of the contents when a pulling force is applied at the center of each side seal. This force shall be applied by holding each side seal between thumb and forefinger of each hand, while simultaneously exerting a slight pull with both hands. Any evidence of loss of vacuum shall be classified as a major defect. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be I and the AQL, expressed in terms of defects per hundred units, shall be 0.65.

4.2.5 Pouch closure seal strength testing. Eight filled and sealed pouches shall be selected at random from the lot. The filled and sealed pouches shall be tested for closure seal strength in accordance with the procedures specified in ASTM F 88, except that the testing speed may be 10 or 12 inches per minute. Three adjacent specimens, 1/2 or 1 inch wide, shall be cut from the top closure seal of each pouch in the sample. The results shall be reported to the nearest 0.1 pound per linear inch. The average seal strength of the closure seal shall be calculated by averaging the test results of the three test specimens cut from the seal. Any test specimen or average seal strength failing to meet the requirements specified in 3.2 shall be classified as a major defect and shall be cause for rejection of the lot.

4.2.6 Meal bag examination. The filled and sealed meal bags shall be examined for the defects listed in table IV. The lot size shall be expressed in bags. The inspection level shall be I, and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 4.0 for minor defects. The sample meal bags shall be selected from shipping containers that have been filled and closed.

TABLE IV. Meal bag defects

Category		Defect
Major	Minor	
101		Not clean - the meal bag or any of its contents 1/
102		Stack height for complete meal not as specified
103		Meal component missing
104		Incorrect assortment for meal package
105		Not arranged as specified
106		Meal bag not close-fitting

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TABLE IV. Meal bag defects (cont'd)

Category		Defect
Major	Minor	
107		Excess air remaining in bag after final closure
108		Not heat seal closure
109		Labeling missing, incorrect, or illegible
110		Loss of vacuum in individual component pouch 2/
	201	Tear, hole, or open seal in meal bag

1/ Outer packaging shall be free from foreign matter which is unwholesome, has the potential to cause pouch damage (for example, glass, metal filings, etc.), or generally detracts from the clean appearance of the package. The following examples shall not be scored as defects for unclean:

a. Foreign matter which presents no health hazard or potential pouch damage and which can be readily removed by gently shaking the package or by gently brushing the package with a clean dry cloth.

b. Dried product which effects less than 1/8 of the total surface area of one pouch face (localized and aggregate).

c. Water spots.

2/ When vacuum retention cannot be determined by obvious cling of the pouch walls to the contents, retention shall be verified by examination as specified in 4.2.4.

4.2.7 Shipping container examination (for assembled meals). Shipping containers shall be examined for defects in assembly, closure, and reinforcement (when applicable) in accordance with the examination criteria of the appendix to PPP-B-636. In addition, the following defects shall be included in the table of examination:

Major - National stock number, item description, contract number, or date of pack markings missing, incorrect, or illegible
- Compartments not as specified
- Number and arrangement of contents not as specified

Minor - Other required markings missing, incorrect, or illegible
- Not a snug-fitting container

4.2.8 Unit load examination. Unit loads shall be examined in accordance with the quality assurance provisions of MIL-L-35078.

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5. PACKAGING

5.1 Preservation. Preservation shall be level A.

5.1.1 Level A.

5.1.1.1 Unit packing. Each complete meal (see 3.5) shall be unit packed in a close-fitting polyethylene bag conforming to Type II, Style 1 of PPP-B-26. The pouches of powdered product shall be arranged in a stack three to five high, as applicable, for each meal within the polyethylene bag. The polyethylene meal bag material shall be 0.003 inches thick. Excess air shall be expelled from the polyethylene meal bag just prior to closing to maintain a close fit between the bag and contents. Each meal bag shall be closed by heat sealing as specified in PPP-B-26.

5.2 Packing. Packing shall be level B or C as specified (see 6.2).

5.2.1 Level B packing. Twenty-seven meal bags, three meals each of breakfast, lunch, and dinner for three days, shall be packed in an upright position in a snug-fitting fiberboard box conforming to style RSC, grade V3c of PPP-B-636. Each box shall be provided with vertically fluted, full height fiberboard partitions forming four compartments with two compartments measuring approximately 12-5/8 inches by 6 inches and the remaining two compartments each measuring approximately 5-3/4 inches by 6 inches. The approximate inside dimensions of the fiberboard box shall be 15-1/2 inches in length, 11 inches in width, and 7 inches in depth. Approximate dimensions are furnished as a guide only. The meals shall be arranged according to days. Day one and day three shall be arranged side by side in the two larger compartments and day two arranged in the two smaller compartments. Shipping containers shall be closed and reinforced with nonmetallic strapping or pressure sensitive adhesive filament reinforced tape in accordance with the appendix of PPP-B-636.

5.2.2 Level C packing. Twenty-seven meal bags, three meals each of breakfast, lunch, and dinner for three days shall be packed as specified in 5.2.1, except that the shipping container shall be in accordance with Uniform Freight Classification or National Motor Freight Classification, as applicable, except that the closure of fiberboard boxes shall be in accordance with the appendix of PPP-B-636.

5.3 Packing of components for shipment to meal assembler. Not more than 40 pounds of product of one kind only, unit packed as specified in 3.1.2 and 3.2, shall be packed in a manner to ensure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The shipping container shall be in accordance with the National Motor Freight Classification or Uniform Freight Classification, as applicable, except fiberboard containers shall be closed in accordance with method II, as specified in the appendix of PPP-B-636. When metal fasteners

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are used in the manufacturer's joint or setup of the fiberboard box, the fasteners on the inside of the box shall be covered with tap or paperboard to protect the pouches from mechanical damage.

5.4 Unit loading. When specified (see 6.2), the product, packed as specified in 5.2.1 and 5.2.2, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified. When unit loads are strapped, the strapping shall be limited to nonmetallic strapping.

5.5 Labeling and marking.

5.5.1 Meal bag. Each polyethylene meal bag shall have labeling that complies with the Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder. The label shall be pressure-sensitive and shall be colored a dull nonreflective color other than white. All printing thereon shall be black, boldface, legible type with indelible ink. The label shall be applied to one face of the empty meal bag with sufficient pressure to ensure its adherence to the bag. Labels shall contain the following information:

MEAL, POWDER (FOR ORAL LIQUID FEEDING)

DAY NO. _____ (1, 2, or 3, as applicable)

MENU TYPE _____ (Breakfast, Lunch, or Dinner, as applicable)

COMPONENTS _____ (List as specified in Table II)

NOTE: TO PROVIDE A NUTRITIONALLY ADEQUATE DIET, EACH MEAL MAY BE SUPPLEMENTED WITH A SNACK OF JUICE, MILK, OR HOT CHOCOLATE. COFFEE OR TEA MAY ALSO BE PROVIDED.

Name and address of assembler

5.5.1.1 Alternative labeling and marking.

a. If a label is placed inside the polyethylene meal bag, between the meal bag and a component pouch, it shall comply with the requirements of 5.5.1 but shall be free of adhesive.

b. The above information (5.5.1) may be printed directly on the polyethylene meal bag with indelible black ink. When printing is applied directly to the polyethylene meal pouch, it shall be located in an area that will permit the printing to be legible after sealing.

5.6 Shipping containers. In addition to any special marking required by the contract or order, containers shall be marked in accordance with MIL-STD-129.

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5.7 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078.

6. NOTES

6.1 Intended use. The products are intended to be used by patients requiring an oral liquid or pureed diet (e.g., those with maxillofacial injuries, wired jaws, oral surgery, dental diseases, or swallowing difficulties).

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Level of packing required (see 5.2).
d. Type and class of unit load when unit loading is required (see 5.4).

6.3 Subject term (key word) listing.

Combat field feeding
Hospital feeding
Jaw injury
Liquid diet
Operational ration
Pureed
Shelf stable

6.4 Changes from previous issue. Marginal notations are not used in this revision due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - SA
Air Force - 50

Preparing activity:

Army - GL

(Project 8970-A150)

Review activities:

Army - MD, QM
Navy - MC
DP - SS

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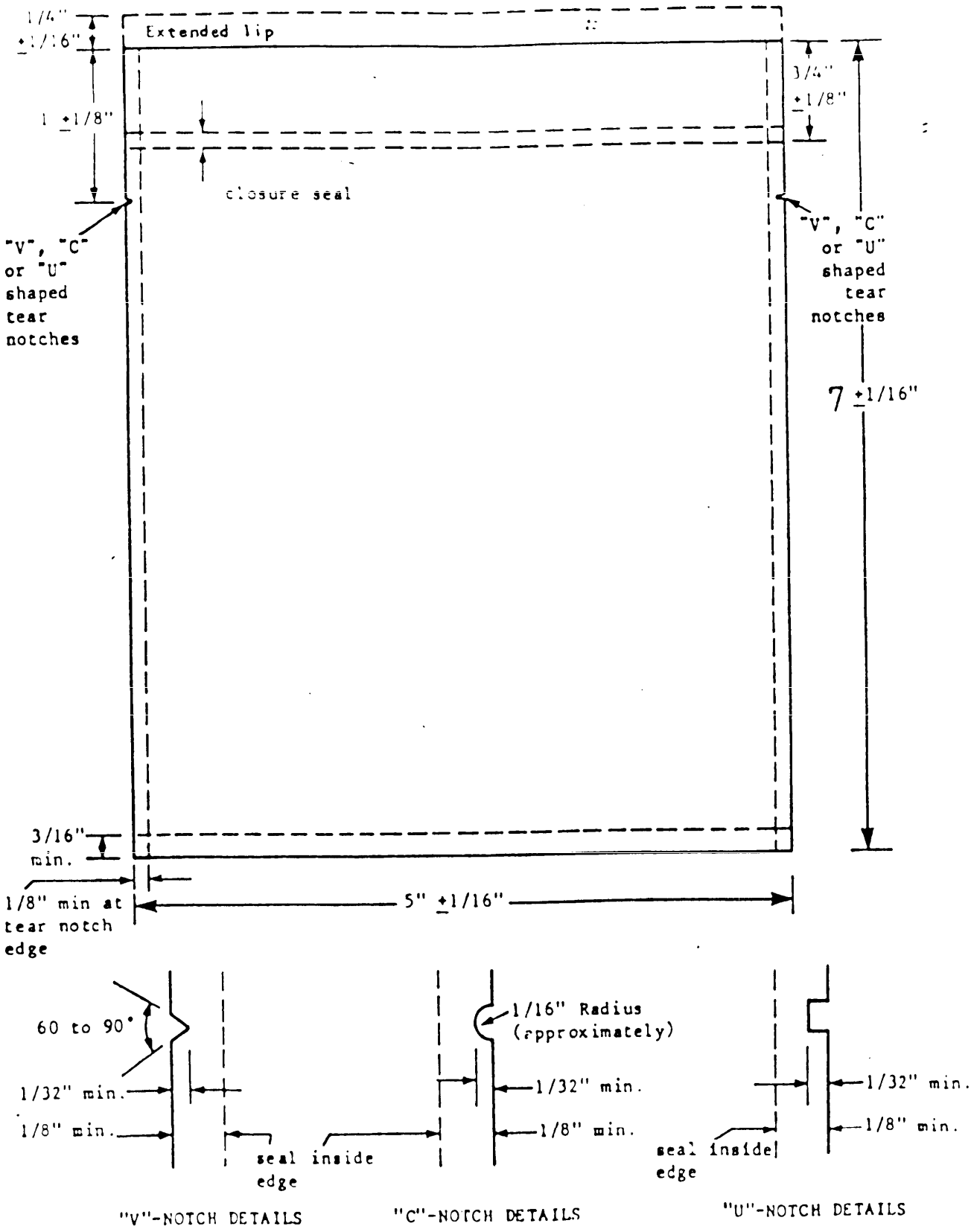


Figure 1. Pouch

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-P-44151A	1992 September 30

3. DOCUMENT TITLE
PACKAGING INTO POUCHES AND MEAL ASSEMBLY OF ORAL LIQUID FEEDING POWDERS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S. Army Natick RD&E Center	(1) Commercial 508-651-4501	(2) AUTOVON/DSN 256-4501
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: SATNC-WTP Natick, MA 01760-5018	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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